LAB 9: Component Diagram

Smart Waste Collection Management System

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1. Objective

To draw UML component diagrams of the Smart Waste Collection Management System showing its main software and hardware components, the interfaces they provide and require, and how they are connected. Diagram 1 shows the whole system; Diagram 2 shows the internal modules of the backend server. The design follows SRS v1.0 (sections 2.1, 2.4, 3.2 to 3.4 and 5.4).

2. Component diagrams included

Figure	Scope	SRS reference
Diagram 1 (page 2)	System level: sensor node, backend server, admin dashboard, driver app, citizen app, database, mapping API and push service.	2.1, 2.4, 3.2 to 3.4
Diagram 2 (page 3)	Backend internals: REST API layer, auth, ingestion, alert, route optimization, complaint and analytics services, adapters and data access layer.	3.3, 5.4 (modular design), REQ-BM, REQ-AL, REQ-RO, REQ-DB, REQ-CF

3. Components and interfaces (system level)

Component	Provides	Requires
Sensor Node Firmware (ESP32 / Arduino)	-	IDataIngestion (HTTP / MQTT)
Backend Server	IDataIngestion, IDashboardAPI, IRouteAPI, IComplaintAPI	IStorage, IMapping, IPush
Admin Web Dashboard	-	IDashboardAPI (HTTPS / JSON)
Driver Mobile App	-	IRouteAPI (HTTPS / JSON)
Citizen Mobile App / Web Form	-	IComplaintAPI (HTTPS / JSON)
Database (MongoDB / PostgreSQL)	IStorage	-
Mapping API (Google Maps / OSM)	IMapping	-
Push Notification Service (FCM)	IPush	-

4. Notation used

Symbol	Meaning
Rectangle with component icon and <<component>>	A software or hardware component (blue = client, green = system, grey = data / external service)
Ball (circle) on a line	Provided interface - a service the component offers
Socket (half circle) around the ball	Required interface - the component uses that service
Dashed green frame	Boundary of a larger component that contains the modules

Component Diagram 1 - Smart Waste Collection Management System

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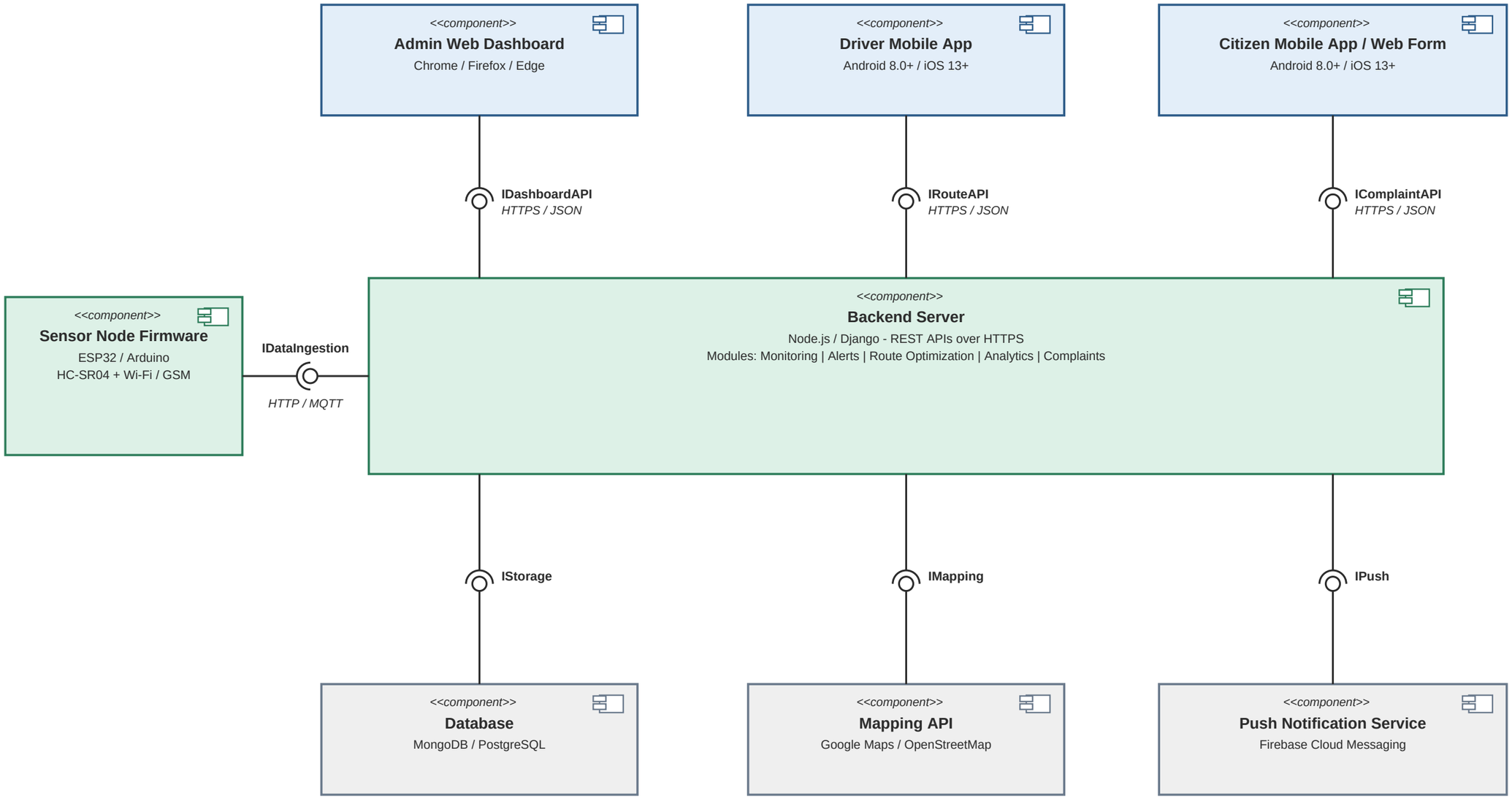

Legend

- Provided interface (offered by the component)
- Required interface (used by the component)

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Component Diagram 2 - Backend Server (Internal Modules)

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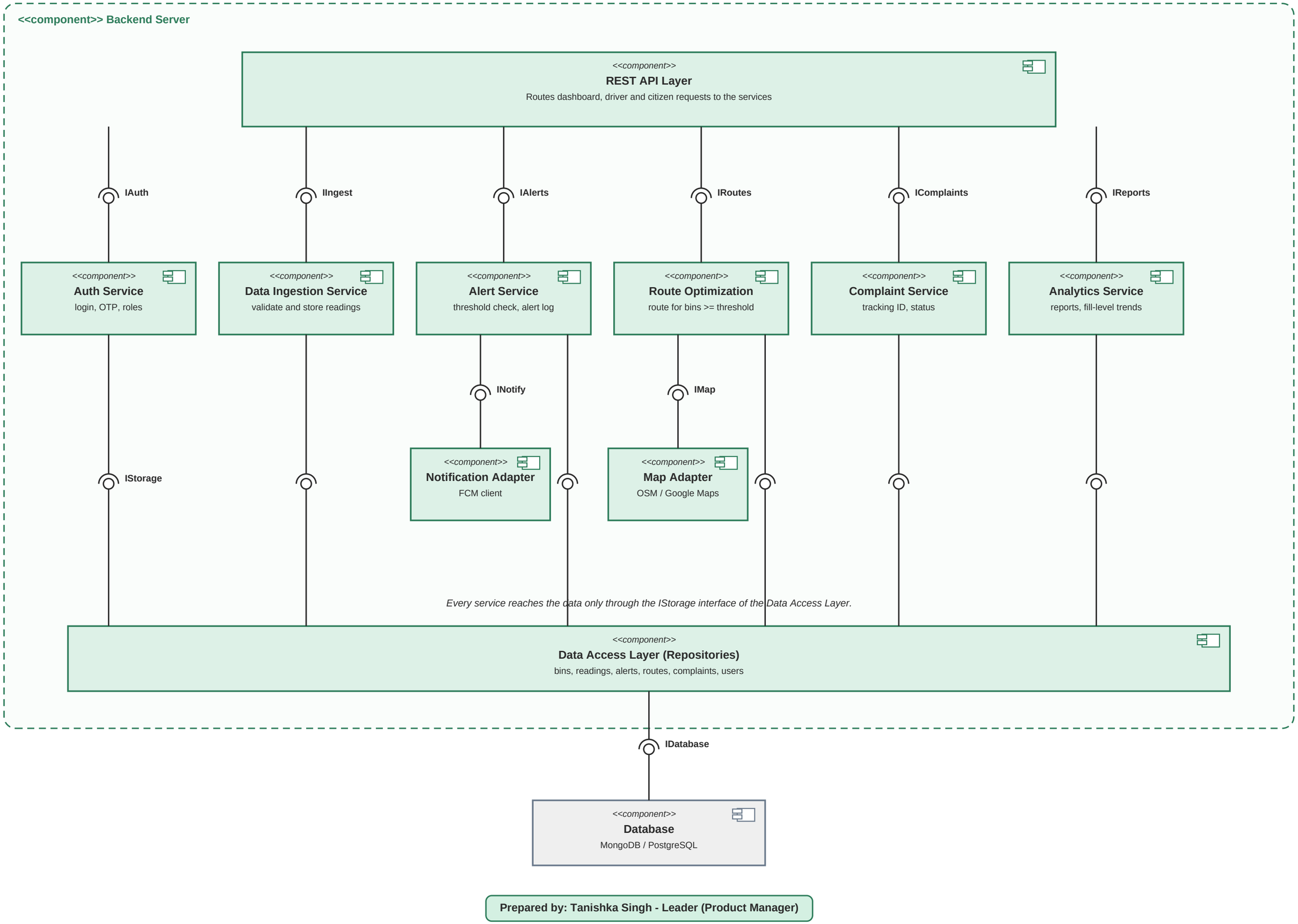

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